No Kid Is an Island: Intergenerational Mobility and Peer Effects

Preliminary draft, please do not circulate.

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Abstract

This paper investigates the implications of exposure to peers from different family backgrounds on children's intergenerational mobility. Using administrative data on Danish high school students and idiosyncratic within-school variation in cohort composition, I find that that a one standard deviation increase in average schoolmates' parental earnings results in a 3.6% increase in lifetime earnings for low SES children. The magnitude of the effect is U-shaped with respect to own parental earnings. When exposed to better peers, low SES children increase their educational achievements and improve their employment outcomes, while high SES children increase the probability of becoming the coworker of a schoolmate. To assess the relative importance of the two mechanisms, I estimate a model allowing for peers' parental background to affect both human capital and access to job opportunities. I find that spillovers in human capital formation are offset by increased competition from schoolmates' higher human capital, while network advantages inherited from former schoolmates explain most of the effect. Overall, these results provide evidence on the joint nature of social mobility: children climb the earnings' ladder with respect of their parents, and in doing so, their peers play a crucial role.

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1 Introduction

Parental income and investments are crucial for children's human capital formation and skills' development (Becker and Tomes, 1979; Cunha and Heckman, 2007; Carneiro et al., 2021). At the same time, as members of neighborhoods and schools, children influence each other through peer effects (Epple and Romano, 2011; Sacerdote, 2011). As a consequence, the connection between exposure to peers and transmission of inequalities across generations might play a crucial role in determining access to opportunities.

This hypothesis is supported by several facts. First, families vaule high-achiving peers when choosing neighborhoods (Eshaghnia et al., 2023) and schools (Abdulkadiroglu et al., 2020) for their kids. Second, conditional on parental earnings, children experience significant heterogeneity in access to opportunities across different neighborhoods (Chetty and Hendren, 2018; Chetty et al., 2016; Chyn, 2018[1]). Finally, Chetty et al., 2022b, documents a strong association between social connectedness to high-SES individuals and upward economic mobility.

In this paper, I test whether and how, conditional on parental characteristics, early exposure to peers from different parental backgrounds affects lifetime economic outcomes. In doing so, I account for the process of intergenerational transmission of inequalities being interdependent across households: each child is influenced by her parents, her peers, and potentially peers' parents too. Leveraging on administrative data from Denmark, I focus on the impact of high school schoolmates' parental background on lifetime earnings and I investigate whether this exposure effect exists and through which channel is at play.

While the hypothesis of parents anticipating peer effects for their offsprings when choosing neighborhoods and schools has been formulated in the literature in different circumstances[2], different causal mechanisms might determine such influence. For example, a child might be affected by spillovers in human capital formation as a consequence of being schooled with schoolmates from higher earnings families. This might be the case if high-SES schoolmates are dispropotionately higher-achieving peers or better role models. Moreover, high-SES peers might provide an advantage because

[1] Mogstad and Torsvik, 2021 or Chyn and Katz, 2021 for an overview of the literature.

[2] Among others, see the notion of *purposive sorting* in Heckman and Landersø, 2022 and *membership theory of inequality* in Durlauf, 1997.

of thier (or thier parents') connections resulting in more valuable referrals or employment opportunities on the labor market. Finally, children from different parental backgrounds might be affected differently from each of the channles outlined above.

To identify exposure effects and their determinants, I use administrative data on the universe of Danish students who enrolled in high school between 1998 and 2004. For those individuals, I document two motivating facts: significant intergenerational persistence in pre-tax earnings and school segregation by parental earnings. The rank-rank correlation between parents' and children's earnings is 0.2. Meanwhile, children from the lowest (highest) quartile of the national earnings distribution have a 16% (32%) higher probability of being schooled with children from the same parental background than if their school allocation were independent of parental earnings. Overall, Children's *inherit* their parents' earning levels and do so surrounded by peers experiencing similar family backgrounds. Therefore, if peer effects are at play, the effect of parental earnings on offsprings' earning potential might be amplified by the exposure to schoolmates who are experiencing a similar parental background, with potential consequences for transmission of inequalities and access to opportunities.

To identify the causal impact of exposure to peers, I compare students who attended the same high school in different cohorts. In doing so, I implement a within-school across-cohorts design (Hoxby, 2000; Black et al., 2013; Carrell et al., 2018; Brenøe and Zölitz, 2020). This addresses the primary concern for identification: endogenous sorting into peer groups. The main assumption is that cohort-specific deviations from the time trend of average school composition are uncorrelated to unobserved factors affecting student outcomes. I argue that this assumption is sustainable since, due to lack of coordination in timing of births and relevants costs in manipulating entry into school parents cannot neither forecast neither react to cohort specific deveiations. To support this assumption, I document that cohort-specific deviations in average school composition are (*i*) uncorrelated with children's parental earnings, (*ii*) uncorrelated with predetermined characteristics such as parental education or parental employment, (*iii*) uncorrelated over time.

A further concern revolves around the possibility of school specific shocks being potentially correlated with school composition and individual outcomes. An example of such a shock would be a change in school endowments or policies. However, I find that deviations from the school trends experienced by placebo cohorts who attended the same school in adjacent cohorts do not affect the outcome of interest. Alternatively,

one might be concerned that the rersults might be biased by unobserved correlated shocks to the local economic environment affecting both parents'and childrens earnings. However, the robustness of the results to the inclusion of cohort-by-municipality fixed effects makes it unlikely for fluctuations in local economic conditions to be driving the results. Finally, I argue that it is unlikely that families have perfect foresight on cohort-specific deviations from secular trends in determinants of school effectiveness (e.g. teacher hiring, infrastructure) before enrollment. This assumption is validated by the estimates not being affected from the inclusion of highly nonlinear school-specific time trend and other school observables such as cohort size.

To decompose the role of spillovers in human capital from network effects on the labor market, I develop a model encompassing both processes. Resuming to a linear-in-means model as in Manski, 1993 allows me to express the reduced form estimate of the impact of peers' parental background as the sum of four different mechanisms. Initially, Earnings are affected by increases in human capital due to *spillovers while in school*. Upon employment, earnings are influenced by own and peers' human capital levels (*spillovers after school*), by peers' success (*schoolmates' network*) and own parents' and peers' parents' (*schoolmates' parents' network*) influence on the labor market.

To achieve identification of the model, endogenous sorting and simoultaneity issues extensively explored in the literature (Manski, 1993) must be addressed. Accounting for school-specific time trends allows me to deal with endogenous sorting reliying on the same assumption outlined above. However, I am still left with the impossibility to separately identify the impact of a peer's characteristics (parental earnings or human capital) from the impact of the same peer achieving better outcomes (education or earnings), since the former are conducive of the latter[3].

To distinguish the specific impact of former schoolmates' labor market achievements from exposure to their human capital, I leverage variations in earnings due to former schoolmates' coworkers' wages. This approach rests on the assumption that coworkers' wages (*i*) influence the earnings of an individual and (*ii*) only impact former schoolmates of the same individual via improvements in that individual's wage. Consequently, the earnings of former schoolmates' coworkers serve as a valid instrument to isolate

[3]As far as human capital accumulation is concerned, I do not distinguish the two effects. Thus, *spillovers while in school* are estimated as the combined influence on educational achievement due to direct exposure to higher-SES parents and to their children, who achieve higher levels of education themselves.

the effects of changes in peers' employment outcomes, given their human capital level. By employing such an exclusion restriction to identify endogenous peer effects, my methodology can be conceptualized as a partial population experiment (Moffitt, 2001) exploiting the partial non-overlapping of the networks of schoolmates and coworkers (Bramoullé et al., 2009; De Giorgi et al., 2010).

Finally, to distinguish the impact of parental earnings due to human capital from that due to parental network effects, I exploit variation in parental earnings after children completed education as driven by parents' coworkers. I argue that this is the relevant variation: conditional on two parents having access to the same earnings while their kids are in school, I identify the impact of them being linked to coworkers who fare better (or worse) when their kids are on the labor market themselves.

I find that, conditional on parental background, a one percentile increase in schoolmates' parental earnings results in a 0.12 percentiles' increase in earnings between the age of 28 and 32. This accounts for two thirds of the residual correlation between childrens' and parental earnings. Notably, I find significant heterogeneity of exposure effects, with the effect being twice as large as the averge for children of parents from the bottom quaritle of the earnings' distributions. The magnitude of these effects is such that increasing the average schoolmates' parental background by one standard deviation ($\sim$ 10 percentiles) for children from low-SES families, increase their earnings by 2.3 percentiles, resulting in a 3.6% change in nominal terms. Moreover, the magnitude of the effect is U-shaped with respect to parental income. The minimum effect is measured for children from parents at the third quartile, which is not statistically different from zero, while children at the top quartile experience the same effect as the average of the sample.

Subsequently, I document the impact of being exposed to peers from more affluent families on education and labor market outcomes. Low SES children exposed to a one standard deviation increase in average schoolmates' parental earnings increase the probability of obtaining a college degree by 1.3*pp*, the probability of being employed in managerial occupations by 2*pp* and the firm AKM fixed effect of the firm they are employed at the age of 30 by 0.7%. For these outcomes, I document that the effect is decreasing in parental earnings, with the effect being zero or negative for high SES children.

The effect of the same esposure on the probability of joining a firm where a high-

SES schoolmate (or a schoolmates parent) is U-shaped in parental background. When exposed to one standard deviation increase in peer quality low-SES (high-SES) child increase the probability of becoming the coworker of a high-SES schoolmate is employed by 1*pp* (0.5*pp*) and the probability of joining a plant of a high-SES schoolmate's parent by 2*pp* (1*pp*).

When decomposing the average effect[4] of exposing children to schoolmates from different parental backgrounds, spillovers in human capital formation while in school and network effects from former schoolmates' parents are found to be the main drivers of the mechanism. Namely, spillovers in human capital formation while in school account for 157% of the effect. However, this mechanism is then combined with two opposing forces: the higher educational achievemnts of schoolmates decreases earnings (–252% of the main effect) while positive network effects due to being in touch with former schoolmates and their parents when navigating the labor market account for 34% and 163% of the estimated exposure effect, respectively.

This paper contributes to the literature of intergenerational mobility and neighborhood effects by proposing and testing peer effects as a link between households sorting and children's access to economic opportunities. In doing so, I provide empirical evidence of localized spillovers in education (Benabou, 1993 and Durlauf, 1997) which link students' human capital accumulation processes to those of their neighbors. The findings in this paper are consistent with previous results from Chetty et al., 2011 on peer effects in kindergarten from the STAR project, as well as more recent investigations by Abdulkadiroglu et al., 2020 and Eshaghnia et al., 2023, which highlight parents' consideration of peers when selecting neighborhoods and schools for their children. Furthermore, this paper presents evidence on the causal impact of early exposure to peers from higher socioeconomic backgrounds, consistent with findings from Chetty et al., 2022a, which indicate that a greater proportion of high socioeconomic status Facebook friendships for those from lower socioeconomic backgrounds is associated with increased social mobility.

A separate contribution lies in the identification of the different roles played by spillovers in human capital accumulation and network ties on the labor market in determining the role of peer effects for social mobility. In this direction, my findings build upon the work of Rothstein, 2019, who suggests that a significant portion of the variation in

[4]More work is ongoing to estimate this decomposition allowing for heterogeneity of effects based on parental background.

access to opportunities across US neighborhoods cannot be attributed to differences in school quality alone and suggests it might be explained job networks and local labor markets institutions. Moreover, my results are complementary to those of Dobbin and Zohar, 2023 whose examination of intergenerational mobility in Israel underscores the role of parental background in granting access to higher paying firms. The joint investigation of human capital spillovers and network ties as determinants of social mobility establishes a connection with the research by Cattan et al., 2022, who, looking at norwegian high school students, find that exposure to peers with higher proportions of attendees from elite universities enhances the likelihood of attending similar institutions.

Finally, this paper adds to the expanding literature on peer effects by providing evince of the significance of exposure to peers during high school for long-term economic outcomes, including investments in higher education and labor market achievements. The findings presented complement existing research on the role of schoolmates for educational achievement (Alvarez Marinelli et al., 2023, Fruehwirth and Gagete-Miranda, 2019) and labor market outcomes (Kramarz and Skans, 2014). Furthermore, the results contribute to the emerging understanding of the heterogeneity of peer effects concerning individual characteristics, such as parental background (Fruehwirth, 2013; Feld and Zölitz, 2017; Tincani, 2024). Specifically, my findings suggest that the influence of peers' parental background on lifetime earnings follows a U-shaped pattern concerning socioeconomic status.

From a methodological perspective, I construct and evaluate a linear-in-means peer effects model, (based on Manski, 1993), enabling a structural interpretation of the reduced form parameter derived from variations in peer composition across cohorts within the same school. This approach allows for separate identification of different components of peer effects: human capital spillovers and enhanced position in labor market network. In doing so I build on the notion of partial population experiment to identify endogenous peer effect introduced by Moffitt, 2001 and I perform an exercise similar to (Angelucci and De Giorgi, 2009, Rossi and Xiao, 2023) wherein variation in the probability of exposure to exogenous shocks across groups allows for identification of endogenous peer effects.

Disentangling of the role of human capital spillovers and network advantages contributes to the discussion on the determinants of inequalities due to participation in different peer groups in two different directions. On one side, light is shed on the mech-

anism trhough which inequalities are replicated by differential exposure to peers. On the other side it highlights the importance of considering both schools and labor maket as playing a similarly important, but distinguished, role in access to opportunities.

The rest of the paper is organized as follows: section 2 presents descriptive statistics and the institutional framework, section 3 presents the reduced form identification design, section 4 presents the reduced form results, section 5 discusses the semi-structural model, section 6 prensents the estimation results and section 7 discusses the results while 8 concludes.

2 Sample and Institutional Framework

2.1 Sample Selection

Danish administrative registers covering the entire Danish population from 1980 to 2019 are the primary data source of this paper. The sample construction relies on linking children to their respective parental backgrounds. To this end, demographic details, including family linkages links, are extracted from the register of households and families (FAIN). Subsequently, each individual is matched with their occupational history, obtained from matched employee-employer data (IDAN and IDAP), as well as their lifetime earnings profile from tax registers (IND). Additionally, information about educational attainment and school attendance is derived from the register of education (UDDA). All the data contained within these registers is collected annually by the central statistical office of Denmark (DST).

The main sample used in this study includes individuals who enrolled in high school between 1998 and 2004. This specific time window is chosen to maximize the probability of observing earnings for each individual by the age of 32[5]. I exclude later cohorts due to potential fluctuations in earnings throughout the life cycle which may arise from ongoing educational pursuits during the latter part of the twenties, and could potentially impact the estimation of children's lifetime earnings. Within the designated cohorts of interest, the registers record a total of 345, 273 children[6]. Out of those, 274 who lack parental linkage are excluded. Furthermore, 19, 963 children who are not present in the registers upon reaching the age of 28, and 48 cases with missing data on parental

[5]The modal student enrolls in highschool in the year she turns 17 and I have data on earnings until 2019.

[6]I exclude students who enroll in hihghschool after the age of 20.

earnings, are also excluded. Finally, I exclude 218 children whose school-cohort is a singleton and 3,265 whose school is reported for less than 4 consecutive years. The final sample consists of 321,511 children, who attended 314 schools in 7 cohorts. The main characteristics of the sample are summarized in Table 1, where the upper panel provides information on individual students and the bottom panel illustrates group characteristics.

[Table 1 HERE]

2.2 Variables' Description

Individual Earnings I measure earnings as the sum of income from the main occupation and self-employment before taxation (*erhvervsindk* in IDAN). I focus on the average earnings between the ages of 28 and 32. By selecting this specific period, I exclude the years when individuals are potentially in education, thereby minimizing potential distortions arising from variations in life-cycle earnings profiles. Instead of relying on nominal earnings values, I use the percentile ranking of the average earnings measure relative to the distribution of children from the same cohort. By doing so, I abstract from changes in the variance of the income distribution over time, ensuring that the estimates are not influenced by changes in income inequality.

Parental Earnings Parental earnings are measured as the percentile of the average of earnings over over the first 18 years of life of the kid relative to the distribution of parents of children from the same cohort in the sample.

School-Cohort I group children by the first high school (*igninstno* in UDDA) they enrolled in and the year in which they did so.

Education I use information on the highest level of completed education (*hfaudd* in UDDA) to compute years of schooling.

Neighborhood In Denmark, parishes are administrative divisions that are subsets of Danish municipalities. Each individual is geo-located based on the parish of residence in the year they complete 10th grade. I use the term parish and neighborhood interchangeably in the remaninder of the paper.

Occupation I classify occupations by sector of employment (*diskokode* in IDAN). Children are classified accordingly to the last occupation recorded at age 30.

2.3 Institutional Setting: Danish High-Schools

Danish students complete compulsory education by age 16 at grade 9th. After 9th grade, students enroll in high-school[7]. Children are allowed to enroll any high-school of their choice, conditional on having successfully completed primary education. The main difference among high-schools is wether their curriculum qualify for access to university after completion. Depending on the type of high-school attended (different tracks are available covering alternatively humanities, natural science and social science, or business and socio-economic disciplines, or technological and scientific subjects or vocational training) lenght of the program might range from 2 to 3 years. Upon enrollment in high school, students are grouped into classes fixed for the whole duration of the progam. Students attend the core courses (e.g.: Danish and Mathematics) in those classes, while they form different classes to attend eligile courses (e.g.: French and IT). To abstract from issues arising from endogenous sorting among classes and eligible courses, I define schoolmates who enrolled at the same school in the cam cohort as the relevant peer group.

[Figure 1 HERE]

[Figure 2 HERE]

Figure 1 documents the segregation of childre into peer groups based on parental earnings. The graph plots the probability of a peer sharing the same parental background as a kid from each decile of the distribution of parental earnings. If kids were allocated to school-cohorts independently of their family background this probability would be 0.1. The figure can thus be interpreted as evidence of extensive sorting of kids across peers' groups, especially at the top and at the bottom of the distribution. While different mechanism could generate this finding, this evidence is consistent with the notion "purposive sorting" already documented for primary schools in Denmark by Heckman and Landersø (2022) wherein better-endowed parents tend to concentrate in the same schools, leading to a correlation between the income levels of children's parents and the parents of their schoolmates. The same plot depicts the degree of sorting by neighborhood and year of birth, suggesting a similar degree of sorting. This lat-

[7] Stuednts have the option of attending an extra 10th grade before enrolling in highschool.

ter comparison is relevant since, in an insitutional context like Denmark where most high-schools have no tuition fees, sorting of households across neighborhoods is expected to play a role in shaping the relation between parental erning levels and school enrollment.

Figure 2 presents observational evidence of the result of this institutional arrangement for inter-generational mobility, suggesting two key mechanisms. First, as appears from panel A, where average earnings for children are plotted by percentile of parental earnings, children from higher-income parents tend to have higher incomes themselves. Second, as appears from panel B, where the average of schoolmates earnings is plotted by decile of parental earnings, children from higher-income parents are also exposed to schoolmates whose parents have higher income levels.

On one hand, this challenges the disentanglement of the effect of parental income from the effect of being exposed to better schoolmates' parents. On the other hand, if exposure to peers outside the family (as schoolmates and schoolmates' parents) matters, this level of segregation would result in children from higher-income families enjoying a double advantage: the first from their own family background and the second from the peers they are potentially exposed to. The identification design presented in the next section will overcome the main source of bias due to endogenous selection of children into peer groups, while the discussion of the results will highlight the interplay between the exposure effect and the reinforcement of intergenerational transmission of income levels due to parents' residential and school sorting.

3 Empirical Strategy

As highlighted above, children are not randomly allocated to schools. Thus, it is reasonable to expect that: (*i*) households sort in different schools according to both observable and unobservable characteristics; (*ii*) unobservable characteristics of the children are correlated with parental income. If the two conditions just outlined realize, a naive regression of classmates' parents' earnings on children's outcomes would be biased. Part of the estimated effect would be due to the intragroup correlation of individual unobservable characteristics of the students. In this section, I introduce the identification strategy of this paper: exploiting variations in parental earnings within school and across cohorts results in the identification of the effect of exposure to schoolmates' parents. The identification strategy employed in this study builds on the work of Hoxby

(2000) and is subsequently applied in similar contexts such as Black et al. (2013), Carrell et al. (2018), Deutscher (2020), and Brenøe and Zölitz (2020).

I estimate the following model:

$$Y_i = \beta_0 + \beta_1 X_i + \beta_2 \bar{X}_{-i} + \gamma_{s(i)} + \tau_{s(i)} c(i) + \delta Z_i + \varepsilon_i \tag{1}$$

Y_i and X_i represent earnings (in percentiles) for each child i and her parents, respectively[8]; $\bar{X}_{-i}$ is the leave-one-out mean of parental earnings (in percentiles) across the schoolmates' of i, where I consider as schoolmates those children enrolled in the same school in the same cohort, excluding i. Moreover, $c(i)$, $s(i)$ denote the cohort and the school of individual i, respectively. Hence, $\gamma_{s(i)}$ is a set of school fixed effects, while $\tau_{s(i)} c(i)$ is a school-specific time trend. Finally, Z_i is a vector of predertermined individual characteristics such as gender, year of birth, year of birth of the parents and size of school-cohort.

Taken together, $\gamma_{s(i)}$ and $\tau_{s(i)} c(i)$ control for school-specific time trends.As a consequence, the estimation of the model in equation 1 relies on within-school comparisons and exploits only cohort-specific deviations from school specific secular trends in school parental earnings. The main identifying assumption is that deviations from school-specific time trends in school composition are not correlated with unobservables affecting students' outcomes. Two mechanisms underlay the credibility of this assumption: parents are unable to anticipate the exact timing of birth of potential schoolmates and it is costly for families to adjust the timing of entry of children into school.

3.1 Support of the identifying assumption

This paragraph will introduce a variety of evidence in facor of the claim that cohort specific deviations from school specific time trends in school composition ($\bar{X}_{-i}$) are as good as random from the perspective of the individual students.

First, Table Table 2 presents a comparison of the standard deviation of $\bar{X}_{-i}$ before and after accounting for school-specific time trends, documenting that approximately one-fourth of the variation in average earnings of classmates' parental earnings cannot

[8]Earnings are measured at the age 28 – 32 for children and between age 0 and 18 of theis child for parents.

be predicted by this approach: this is the variation exploited for identification.

[Table 2 HERE]

Second, as clear from Figure 3, the distribution of the residuals of peer group characteristics after accounting for school specific time trends is well approximated by a normal distribution.

[Figure 3 HERE]

Third, student's own parental background is uncorrelated with residuals of peers' parental background. While this statement finds a first confirmation upon inspection of Figure 4, in Table 3 I perform a more careful analysis by regressing the individual parental earninigs on schoomates average parental earnings, alternatively including different specifications for the school time trends and including or omitting individual and school level controls. As noted by Guryan et al., 2009, regressing a variable on the leave-out-average of the same variables induces a mechanical negative bias aven in the absence of peer effects[9]. For this reason, implement the bias correction introduced by the same authors and applied to a similar context by Brenøe and Zölitz, 2020, I include in each regression the leave-one-out average of parental background computed at the school levels. Reassuringly, upon inclusion of school specific time trends one cannot reject the hypothesis that the correlation between own and peers' parental earnings is zero at standard confidence levels. The inclusion of individual and school controls increases the p-value of the test but is not necessary to reach this conclusion and does not alter signifincantly the point estimate.

[Figure 4 HERE]

[Table 3 HERE]

Fourth, students predetermined characteristics are not correlated with residuals of peers' parental background. If agents were to anticipate deviaton from school specific time trends when deciding which school to join we would excpect some degree of correlation between individual characteristics (i.e.: parental education) and the deviation itself. To test this assumption I run OLS regressions of the average peers' parental background on several predertermined individual characteristics, such as demographics, parental education, parental employment and measures of quality of residential ar-

[9] Individuals with the highest realization have, by construction the lower group average.

eas.[10]As summarized in Table 4, upon inclusion of school specific time trends only for 3% of the variables considered the null hypothesis of a zero correlation can be rejected at 99% confidence level. Moreover, among the majority of the correlation reported, and most importantly for all those who are found to be statistically different from zero, the magnitude is negligible[11]. I interpret this as supportive evidence of the fact that residuals in school comoposition are as good as random from the perspective of the students.

[Table 4 HERE]

Fifth, cohort-specific deviations from school time trends are uncorrelated over time. If agents were to use information on the deviations in cohort composition to choose their school, we would expect a degree of correlation in the time series of the residuals, with schools who randomly obtained a better (worse) cohort composition in time t having a better (worse) composition in time $t+1$. In Table 6 I present evidence from school-specific time series regression testing for autocorrelation in cohort composition. Upon inclusion of linear time trends, only 4.5% of the schools in the sample exhibit a correlation over time in school composition statistically different from zero at 90% confidence level.

[Table 6 HERE]

4 Results

[Table 7 HERE]

Column (1) of Table 7 reports the coefficient of a regression of the rank of parental earnings while the kid is between 0 and 18 years old on rank of earnings between the age of 28 and 32. The estimated coefficient suggests that an increase in parental earnings from the 25th to the 75th percentile is associated with an increase of 10 percentiles in their children's earnings. Considenring that the average kid from parents at the 25th percentile ranks at the 46th percentile by the age of 28-32, this is equivalent to increasing her earnings by 13.5% from 40, 000$ to 45, 400$[12]. Column (2) includes in the regres-

[10]The complete list of variables with the respectiev coefficients is reported in Table 5.

[11]Which is, below 0.01. Note that all variables haev been standardized before performing the test.

[12]In compliance with DST regulations, the earning levels reported are not the actual percentiles. The amount reported as earnings for perentile c is the average earnings among all the individuals between percentile c and $c-1$, rounded to the closest hundreds.

sion the leave-one-out average parental earnings among the schoolmates. Column (3) replicates the same regression including cohort and school fixed-effects and columns (4) includes school-scpecific linear time trends.

The coefficient estimated on schoolmates' parental earnings in column (2) indicates a positive and significant relationship between classmates' parental earnings and children's own earnings. The coefficient mantains its sign unchanged upon inclusion of school and cohort fixed in column (3) and school specific time trends in column (4). I interpret the decrease in its size from column (3) to column (4), togheter with the evidence presented in the previous paragraph of residuals being uncorrelated to individual characteristcs upon inclusion of time trends, as potentially induced by changes in expectations of school composition over the time considered. In light of the same argument argument, my preferred specification is that of column (4)[13]. In this specification, the coefficient of interest is positive and statistically different from zero at 95% confidence level. A natural benchmark to interpret the magnitude of this coefficient is to compare it with the correlation in earnings between children and their own parents: exposing children to higher earnings classmates' parents has an effect as large as $\sim$ 65% of the correlation in earnings between parents and children.

A second perspective to appreciate the economic magnitude of the coefficient is from the perspective of the child of the household at the 25th percentile of the earnings distribution. This child earns on average $\sim 39,200\$USD/yr$ by the age of 28-32. If exposed to an increase of one standard deviation increase ($\sim$ 10 percentiles) in schoolmates' parental hes earnings would increase to $\sim 40,000 USD/yr$[14]. Assuming the earnings at age 28 – 32 as a proxy for lifetime earnings, this implies a 1.86% increase in lifetime earnings.

A concern in the interpretation of these results is that unobserved school-specific shocks correlated with the cohort-specific deviation in peer composition might be confounding the estimation of the parameter of interest. If school-cohort-specific specific deviations from school time trends were to be correlated with unobserved fac-

[13]Moreover, as reported in Table 8, including non-linear higher-order specifications of the time trends does not qualitatevely affect the coefficient. I interpret this as suggestive evidence that the linear time trends capture the essential variation in the relationship between exposure to schoolmates' parents and children's earnings outcomes.

[14]In compliance with DST regulations, the earning levels reported are not the actual percentiles. The amount reported as earnings for perentile c is the average earnings among all the individuals between percentile c and $c-1$, rounded to the closest hundreds.

tors affecting students outcome (such as improved school resources or local economic conditions), we would expect such deviations affecting also the adjacent cohort unless the unobserved shock was indipendent across adjacent cohorts. This offers a simple placebo test performed in Table 9 where the same model as in equation 1 is estimated, replacing the measure of peer parental background ($\bar{X}_i$) with the average parental earnings of students enrolled in the same school of i, t cohorts after the actual cohort of i: $\tilde{X}_{i,t}$. The deviation from school time trends in adjacent cohorts does not affect the outcome. I interpret this as ruling out potential correlated shocks driving the estimation of peer effects.

[Table 9 HERE]

However, deviations from school-specific time trends in parental earnings may be correlated with other environmental factors which are uncorrelated over time even at the school level, such as fluctuations in local economic conditions at the time of graduation. As shown in the appendix, the inclusion of cohort-by-municipality fixed effects $\gamma_{c(i)\times m(i)}$ (column (5) of Table 8)does not affect significantly the coefficient of interest. This specification restricts the comparison to students who are esposed to the same economic conditions outside their school environment. The rubustness of the coefficient of interest to this specification addresses the concerns of local shocks correlated with shocks to peer composition driving the estimates.

However, potential correlation between deviations from school-specific time trends and, unobservable, time-varying characteristics at the school level is not ruled out. While it is unlikely that a temporary deviation from the average composition of parents in a school immediately affects the quality of the school, it is possible if parents were directly involved in school activities. In such cases, exposure effects estimated in equation 1 would capture the combined effect of being exposed to better schoolmates' parents per se and their implicit impact on school quality. However, given the age of the students and the type of education considered, direct inteventions from parents on factors determining school quality will be assumed to be second order.

4.1 Heterogeneity of the effect across Socioeconomic Status

[Figure 5 HERE]

Figure 5 depicts the heterogeneity of the exposure effect parental background. The

graph reports the point estimate and the 95% confidence intervals of the marginal effect of exposure to schoolmates' parental earnings implied by including in the model from eq. 1 a full interaction between the schoolmates' parental earnings and a set of dummies for each ventile of the distribution of own-parental earnings. The effect is heterogeneous with respect to parental background: low and SES children are affected more than the average child from exposure to peers, while the effect for children from the middle of the parental earnings distribution is smaller or indistinguishable from zero.

It is suggestive to compare this finding with the degree of segregation highlighted by Figure 1. Those tracts of the population which are more segregated (the two extrema of the distribution) are more affected by exposure effects. Moreover, this degree of heterogeneity might be interesting per se if one considers that welfare policies are usually targeted at families at the bottom of the earnings distribution. To appreciate the economic magnitude of the effect, consider a child whose parents rank at the bottom quartile of the earnings distribution[15]. The model predicts that this child earns on average $\sim 38,500 USD/yr$[16] by the age of 28-32. If exposed to an increase of one standard deviation increase ($\sim$ 10 percentiles) in schoolmates' parental hes earnings would increase to $\sim 39,900 USD/yr$. Assuming the earnings at age 28 – 32 as a proxy for lifetime earnings, this implies a 3.6% increase in lifetime earnings.

4.2 Exposure Effect on lifetime outcomes: education and occupation

In this section, I present evidence on the effect of exposure to higher SES peers on education and labor market achievements by the age of 28-32. For each outcome of interest I re-estimate the model from eq. 1 substituing the dependent variable with the relevant variable. The regression model includes a full set of interactions between schoolmates' parents earnings and a set of dummies for each quaritle of the distribution of parental earnings. Thus allowing for the identification of heteronegeous effects along the the distribution of students' family backgrounds. Table Table 10 presents an overview of the variables considered.

[15]The relevant coefficients from a model including th full interaction between the schoolmates' parental earnings and a set of dummies for each quartile of the distribution of own-parental earnings is reported in the Appendix

[16]The earning levels reported are not the actual percentiles. In compliance with DST regulations data on percentiles are confidential. The number I report in the text is the average earnings among all the individuals at a given percentile, rounded to the closest hundreds.

[Table 10 HERE]

[Figure 6 HERE]

The left panel of Figure 6 illustrates the change in the probability of obtaining a University degree due to a one standard deviation increase in schoolmates' parental background. The effect is decreasing with parental background, being indistinguishable from zero for high SES students. The middle panel of Figure 6 plots the effect of improved peers' quality on the AKM firm fixed effect at the firm of employment at the age of 30[17]. While a positively significant effect of exposure on firm rankings is measured for low SES children, the effect is reversed for students from higher parental background. The right panel Figure 6 reports the effect on the probability of being employed as a manager or a high profile employee (orange) when exposed to better peers in high school. Once again, it's magnitude is decreasing with parental background.

Overall, exposure to better peers in highschool increases the probability of obtaining a university degree, being employed at a higher wage firm and with managerial responsabilty dispropotionately more for low SES children than it does for children from higher earnigs parents.

4.3 Exposure Effect and Workplace Inheritance

Before turning to a decomposition of the effect of increase in education levels from the role played by network effects in the nex section, I will present here suggestive evidence on the fact that peer exposure in school affects employment trajectories via network effects. In fact, having classmates from higher-income parents might benefit children once they enter the labor market via referrals or network effects in general. Leveraging on the universe of employer-employee relations observed in the Danish administrative data I collect the employment histories for the children in the sample between the age of 20 and 30 years. For this period, I document 3 main facts. As from Table 11 First, 6% of the kids are employed at least once at the same plant where one of their parent worked while they were in high-school. Second, 17.4% joined a plant were a schoolmate was previously employed. Third, 28.7% joined a plant were one of the schoolmates' parents was employed while they were in school. Note that the choice of time periods with respect to the age the kid guarantees that the employment relation

[17] AKM firm fixed effect is computed as the firm specific fixed effect in a regression of log wages on individual and firm fixed effect on the universe of Danish worker.

of the parent (or the schoolmate's parent) pre-exists the one the kid.

The probability of inheriting a workplace from a parent is increasing in parental earnings, growing from .03 for children from the first quartile to .08 for children from the highest quartile. This observation is consistent with higher paying occupation-workplaces being more likely to be transmitted along dynasties, for example because of their increased attractiveness to children. Conversely, both the probability or joining the plant of a schoolmate and a schoolmate parent increase is constant with respect to respect to parental background. However, when we consider the probability of joining a schoolmate (or a schoolmates' parent) at their plant conditional on the schoolmate being from high SES, the gradient is again positive. High SES children are twice as likely than low SES children to join the plant of a high SES peer or her parent's plant.

[Table 11 HERE]

[Figure 7 HERE]

The above findings are coupled with the fact that, when esposed to higher SES peers, children from the bottom and the top of the earnings' distribution do increase their chances to join the plants of their peers from the most affluent family backgrounds, while less action is decumented for children from the middle of the earnings distribution. This is documented in figure Figure 7, where the increase in the probability of joining schoolmate or a schoolmate parent at their workplace due to a one percentile increase in schoolmates' parental earnings is plotted separately by quartile of parental earnings. Children from low SES families experienca a 1 pp (2.2pp) increase in the probability of joining the firm of a hihg SES peer (parent) when exposed to a one standard deviation increase in peers'quality. Children from high SES families experienca a 0.5 pp (1.1pp) increase in the probability of joining the firm of a hihg SES peer (parent) when exposed to the same variation.

Considered togheter, the evidence presented in this section is suggestive of a potential channel through which peer effects at the high school level might generate the effect observed in the main results of this paper. If one in six students finds employment at a plant where a schoolmate's parent was previously employed, being exposed to schoolmates whose parents are employed at higher paying firms might improve the advantage experienced by the child when entering the job market.

However, as opposed to spillovers in education achievement that affected children

from the middle and low SES, only low and high SES children appear to increase their chances of joining their peers' network when exposed to higher SES peers. This is compatible with two different processes taking place at the same time. Assuming some degree of substitability between parental investments and peer exposure e diminishing marginal returns in the technology of human capital formation, exposure to higher SES schoolmates generates spillovers in the classroom that increase low SES children investments in education (substituting lack of parental investments), while leaves unaffected high SES children (due to already high levels of parental investments). At the same time, exposure to better peers generates opportunities to leverage schoolmates' network on the labor market: after school, individuals decide whether to give referrals for their schoolmates (or their children schoolmates'). The private nature of the referral decision results in children from high-SES families to be the preferred object of such referrals due to homophily or higher investments in human capital from their families. Separating the role of human capital spillovers from network effects is the contribution of the next section.

5 Accounting for Human Capital and Network Effects: a Semi-structural Model

In what follows, I formalize a model to disentangle the role played by spillovers increasing levels of human capital (H_i) and labor market network effects in determining the relation between exposure to peers' parental background while in school ($\bar{X}_i$) and earnings between the age of 28 and 32 (Y_i). The key challenge in doing so is that exposure to schoolmates from more affluent families may result both in increased levels of human capital and improved network position. I assume human capital to be influenced from parental earnings (X_i), peers' parental earnings $\bar{X}_i$ and peers' human capital attainment ($\bar{H}_i$) according to equation (2).

$$H_i = \alpha + \delta X_i + \gamma \bar{X}_i + \beta \bar{H}_i + \varepsilon_i. \tag{2}$$

I assume earnings to be determined by own and peers' human capital levels (H_i and $\bar{H}_i$, respectively), former schoolmates' earnings ($\bar{Y}_i$), parental earnings after school is completed (X_i') and former schoolmates' parental earnings after school is completed ($\bar{X}_i'$) as described by equation (3). Notably, I also allow individual earnings to be affected from coworkers' earnings (Z_i).

$$Y_i = \alpha' + \pi H_i + \sigma \bar{H}_i + \beta' \bar{Y}_i + \lambda X_i' + \phi \bar{X}_i' + \rho Z_i + e_i. \tag{3}$$

Finally, equation (4) describes the relation between parental earnings while in school X_i with those after school completion X_i'. Also, parental earnings are allowed to be influenced from coworkers' earnings (W_i).

$$X_i' = \mu_0 + \mu_1 X_i + \mu W_i + \xi_i \tag{4}$$

Writing closed form equations for the recursive functions (2) and (3), one is left with the reduced form relation:

$$Y_i = \psi_0 + \psi_1 X_i + \psi_2 \bar{X}_i + \psi_3 W_i + \psi_4 \bar{W}_i' + \rho Z_i + \frac{\beta'}{1-\beta'}\rho \bar{Z}_i + \tilde{e}_i \tag{5}$$

with :

$$\psi_1 = \pi\delta + \lambda\mu_1$$
$$\psi_3 = \lambda\omega$$
$$\psi_4 = \frac{\beta'\lambda + \phi}{1-\beta'}\omega$$

The advantage of this setup is that it allows for the reduced form effect of exposure to schoolmates ψ_2 to be decomposed in four different additive components as in equation (6).

$$\psi_2 = \underbrace{\frac{\beta\delta+\gamma}{1-\beta}\pi}_{\substack{\text{Spillovers in } H \\ \text{at School}}} + \underbrace{\frac{\sigma}{1-\beta'}\frac{\delta+\gamma}{1-\beta}}_{\substack{\text{Spillovers in } H \\ \text{after School}}} + \underbrace{\frac{\delta+\gamma}{1-\beta}\pi\frac{\beta'}{1-\beta'}}_{\substack{\text{Network} \\ \text{Effects}}} + \underbrace{\mu_1\left(\frac{\beta'}{1-\beta'}\lambda + \frac{\phi}{1-\beta'}\right)}_{\substack{\text{Parental Network} \\ \text{Effects}}} \tag{6}$$

First, educational achievement is affected by peers' parental background via peer effects (*spillovers while in school*). Second, earning are influenced by both own and fprmer schoolmates' human capital (*spillovers after school*). Third, earnings are also subject to peer effects (*schoolmates' network*) and affected by the labor market position of parents and peers' parents (*schoolmates' parents' network*). A graphical representation of the four different channels is offered in Figure 8.

[Figure 8 HERE]

We expect *spillovers in school* if the process of human capital formation exhibits peer effects. More specifically, this will be the case either if students decision to invest in education is influenced by exposure to their peers parents or by exposure to their schoolmates endogenous effort. Following the established definition of the literature (Manski, 1993), the former is to be interpreted as a direct or exogenous effect, while the latter as an indirect or endogenous effect. I compute *spillovers in school* as the joint combination of the two.

Spillovers in human capital after school might emerge if, upon finishing school and after having completed their education H_i, children were to be affected by peers' human capital independently of former schoolmates' earnings $\bar{Y}_i$. The direction of this effect is to be empirically determined: higher educated former schoolmates might for example redirect individuals to more productive activities, thus increasing their earnings, or might in fact represent increased competition, thus reducing individual earnings.

Finally, network effects would capture how much each child benefits from peers employment success. Intuitively, if schoolmates (schoolmates parents) were to provide refereralls dispropotionately to schoolmates (childrens' schoolmates), we would capture a direct relation between variation in earnings among peers (peers and parents of peers).

5.1 Identification

To estimate the decomposition in equation (6), three main identification issues must be faced: endogenous sorting, simoultaneity of human capital and peers' earnings and correlation between parental earnings while in school and afterwards.

First, as in the reduced form case above, endogenous sorting into peer groups might bias the results. To address this concern, I estimate the whole model replacing actual variables with residuals from a set of school-specific time trends. Under the same assumption exploited in Section 3 (agents are unable to anticipate cohort-specific deviation from school trends), this results in considering variation in peers characteristics to be considered as good as random.

Second, peers human capital in equation (5) might affect individual earnings directly or via increases in peers' human capital. To separately identify the effect of more successful former schoolmates on the labor market from exposure to their higher human capital, I exploit variation in earnings due to former schoomates' coworkers' wages.Under the assumption that coworkers' wage affect the wage of a given individual[18] and former schoolmates of the same individual only though improvements in the wage of that individual, former schoolmates' coworkers' earnings serve as a valid instrument to identify the effect of changes in peers employment outcomes, conditional on their level of

[18]peer effects on the workplace are extensively documented in the literature, see for example: Mas and Moretti, 2009 and Cornelissen et al., 2017.

human capital. Exploiting a similar exclusion restriction to identify endogenous peer effects, my approach can interpreted as a partial population experiment Moffitt, 2001. Moreover, relying on peers (coworkers) of peers (former schoomates) I am exploiting the intuition developed by Bramoullé et al. (2009) and De Giorgi et al. (2010).

Third, parental earnings while the kid is on the labor market are a function of parental earnings while the kid was accumulating human capital in school, thus generating potential collinearity between X_i and X_i'. To distinguish the impact of parental earnings on human capital from that on earnings due to parental network effects, I control for parental earnings when children are in school (up to age 25) and use the variation of parental earnings due to parents' coworkers' when children are on the labor market (age 28-32) to infer parental network effects. I argue that this is the relevant variation: conditional on two parents having access to the same earnings while their kids were accumulating human capital, I identify the impact of them being linked to coworkers who fared better (or worse) when their kids are on the labor market themselves.

In what follows, I measure parental earnings while in school and children's earnings as in the reduced form analysis. I measure human capital as the number of years spent in education by age 25. I measure parental earnings after human capital completion as parental earnings between age 28 and 32. I measure coworkers' earnings as the leave one out average of the wages of workers employed at the same plant.

Finally, I show in the appendix that the paramters composing (5) are identified upon estimation of the following system of equations.

$$
\begin{aligned}
Y_i &= \psi_0 + \psi_1 X_i + \psi_2 \bar{X}_i + \rho Z_i + \psi_3 W_i + \psi_4 \bar{W}_i' + \frac{\beta'}{1-\beta'}\rho \bar{Z}_i + \tilde{e}_i \\
\bar{H}_i &= \frac{\alpha}{1-\beta} + \frac{\delta+\gamma}{1-\beta}\bar{X}_i \\
H_i &= \frac{\alpha}{1-\beta} + \delta X_i + \frac{\beta\delta+\gamma}{1-\beta}\bar{X}_i + \tilde{\varepsilon}_i \\
X_i' &= \mu_0 + \mu_1 X_i + \omega W_i + \xi_i
\end{aligned}
$$

And noticing $\psi_1 = \pi\delta + \lambda\mu_1$; $\psi_3 = \lambda\omega$; $\psi_4 = \frac{\beta'\lambda+\phi}{1-\beta'}\omega$.

I estimate the parameters both as non-linear combinations of OLS estimators from estimating each equation separately and via simoultaneous GMM estimation. No qualitative difference is detected across the two methods. I present GMM estimates and

leave OLS estimator to the Appendix.

6 Semistructural Results

[Table 12 HERE]

[Table 13 HERE]

[Figure 9 HERE]

Table 12 reports the parameters identified estimating the system of equations from the last section via GMM where every variable is included as the residual after a set of school specific time trends. The first parameter $\frac{\beta\delta+\gamma}{1-\beta}$ identifies the effect of increasing by one percentile the average schoolmate parental background on the years of education attained. The second captures the same effect on the average years of education of the peer group. As discussed above, those two parameters do not disentangle the role played by direct effects (γ) from that played by endogenous interactions (β). Turning to the wage determination equation, π measures the private returns from one extra year of education while σ captures the direct effect of peers human capital con earnings not mediated by own human capital and peers' earnings. Further, the estimates reveal a negligible endogenous effect: β' is positive but close to zero and estimated with a large standard error. Turning to the effect of parental network, the effect of own parental connections is not distinguishable from zero, while the impact of peers' parental network is positive. Finally, coworkers are found to influence their peers' earnings, with the impact being as large as 0.49 for childrens' coworkers and 0.24 for parents' coworkers.

Table 13 reports the estimates for the contribution of each of the four mechanisms and the overall exsposure effect as specified in equation (5). Figure 9 plots the same decomposition graphically. The results suggest the existence of substantial spillovers in human capital while at school. However, such beneficial effects are more than offset by the negative spillovers from peers' human capital on the labor market. I argue that this last effect might be due to increased competition faced on the labor market because of enhanced peers' supply of human capital. When including in the anylsis the network effect from own and peers' parents, however, the advantage of being exposed to higher SES peers in highscool is unambigous. Individuals in the sample experienced exposure effects in highschool both because of the endogenous relation of

earnings profile among former schoolmates and, even more substantially because of the advantages offered by schoolmates parents on the labor market.

7 Discussion

The overall picture drawn in this paper views access to opportunities for children as a joint process, taking place both within and among families who experience some degree of interaction (e.g.: sending their kids to the same school). Specifically, upon joining the same high-school, I document children adult earnigs to be affected by thier peers' employment outcomes. I find that being exposed to a set of schoolmates' parents whose average earnings are one standard deviation higher results in a 1.86% increase in lifetime earnings for the average student. Moreover, this effect is U-shaped with respect to parental background, with low SES children experiencing the same effect to be as large as 3.6%. With this regard, I find low SES children to increase thier educational achievement when exposed to better peers while high SES children experence in the probability of joining a peer's workplace. Finally, upon decomposing the drivers of the estimated effect, I find that on average human capital spillovers while in school (which explain 153% of the effect) are offset by increased competition, while network effects from peers and peers parents account for 38% and 152% of the effect, respectively.

While the within-school across cohorts identification strategy illustrated above is designed to deal with endogenous selection and correlated effects, the results should be interpreted keeping in mind two main caveats. First, the effect of direct interactions with schoolmates' parents is not separately identifiable from the effect of schoolmates' parents mediated from their own kids being peers of each other (i.e.: schoolmates). In other words, since parents invest in their kids' human capital, the estimates presented are both compatible with children having an advantage by being the schoolmates of higher earnings families because of the improved interactions with their schoolmates or because of direct interactions with those adults. Second, combining leave-one-out averages and group fixed effects might result in downward biased estimates. As pointed out by Caeyers and Fafchamps (2020) a mechanical result of including both group-level fixed effects and leave-one-out averages as done in the main specification of this paper results in a mechanical negative correlation between the individual outcomes and the group average since by construction the best member of a group has worse peers than the others. However, the implication of this for the results presented is that, if any-

thing, they are underestimating the true parameters of interest.

Guiding the discussion of the results that follows, two main mechanisms could be envisioned behind the effect of exposure to higher income schoolmates' parent on children future outcomes. The first involves an increase in children's human capital due to interactions either within the school or directly with schoolmates' parents. This would happen if being schooled with a child from an higher-earning family results in having a better schoolmate who potentially improves the process of learning (affecting the classroom environment or through peer effects in general[19]) or in obtaining exposure to different adult role models, who could facilitate the kid to navigate higher education tracks by providing guidance or information. This would be in line with the mechanism of Benabou (1993), where a system of neighborhoods is characterized by the fact that the share of high-educated individuals in each location decreases the cost of achieving higher education for everyone else living in the same location. The second potential mechanism implies an improvement in the position of the child in the social network of the local labor market: children who are schooled with higher earnings schoolmates' parents gain a valuable link which they can leverage on when entering the labor market through the referrals that link might provide to potential employers. This would be consistent with the models of network-based job referrals first introduced by Montgomery (1994) and applied to a neighborhood setting by Bayer et al. (2008).

The increase in the probability of obtaining a degree documented in **??** is consistent with the hypothesis of children accumulating more human capital thanks to exposure to better peers. In particoular, I document that low SES kids increase their chances of getting a University degree by 5.3% when exposed to a one-standard-deviation increase in schoolmates' parental background. On the contrary, no such effect is found for high-SES children. It is worth noting here however, that a slight increase in the university ranking conditional on graduation is measured for high-SES kids. This results is consistent with Cattan et al., 2022 who find that in Norway being in class with children of parents graduated at elites institutions boosts the probability of attending the same institution.

However, the net null result on education coupled with a strong exposure effects when eanings are considered motivates the investigation on the network advantage

[19] see Sacerdote (2011) for an overview.

experienced by high SES children. I document that 1 kid out of 5 joins the plant of a schoolmate parent at least once in the early stage of their career. I couple this with evidence of exposure to schoolmate from more affluent families increasing the probability of actually exploiting network advantages by joining a connected plant for high SES and low SES kids but not for children from the middle of the distribution.

Finally I devolop a decomposition of the main effect of exposure to higer SES schoolmates on adult earnings accounting for both human capital spillovers and network effects. Overall[20], I find that evidence of substantial spillovers in human capital accumulation within the school that are off-set by increased competition on the labor market due to peers' increased human capital.

I interpret this evidence as suggestive of spillovers in human capital driving the effects for low SES-children and netowork effects for high-SES children. School-level spillovers in human capital are public goods in nature, thus affecting every member of the group. However, assuming parental investments to be substitues of peer esposure in the human capital formation technology, the returns to such spillovers might be higher for low-SES kids because of lower level of parental investments, while leaves unaffected high SES children (due to already high levels of parental investments). Conversely, the network (and the use individuals make of it) of acquaintances on the labor market is the result of endogenous private decisions. The private nature of the referral decision results in children from high-SES families to be the object of such referrals due to homophily or higher investments in human capital from their families. Separating the role of human capital spillovers from network effects is the contribution of the next section.

The results presented improve our understanding of the close relation between intergenerational mobility and neighborhood effects highlighted in the literature (Chetty et al., 2014) by identifying one channel by which sorting across neighborhoods and schools affect patterns of mobility: children who are exposed to better peers (which might be identified as one of the amenities of better neighborhoods) experience an increase in their human capital and, most likely according to the evidence presented, in preferential access to high-paying occupations.

The evidence collected on the main mechanisms driving those results and the preminent role of network effects circumstantiate the reasons behind the peer effects mea-

[20]More work is ongoing to estimate this decomposition separately by parental background.

sured on earnings. In doing so, I document the relevance of network effects as one of the main driver of the exposure effect. In doing so, I present empirical evidence supporting an hypothesis already formulated in the literature (Rothstein, 2019, Heckman and Landersø, 2022) engaged with the identification of the determinants of geographic heterogeneity in access to opportunities.

Moreover, the heterogeneity in the exoposure effect by parental background motivates a further interest in potential reallocation policies (such as school desegregation) to retrieve the socially optimal allocation of peers. In fact, if the estimates here presented were to be confirmed, the large share of population (children from middle income families) who is virtually unaffected by school composition, could be identified as potential reciever of some of the children from low SES families. However, further research is due in this direction to develop a prediction on such a policy. The role of families, and the degree of complementarity between their direct investment and that coming from peer exposure should be taken into account.

8 Conclusion

This paper has shown that exposure to schoolmates from different parental backgrounds affects children's opportunities for economic mobility. When comparing children who attended the same school in different cohorts, those whose schoolmates' had higher average earnings experienced higher earnings themselves when adults. The size of the effect implies that increasing the average earnings for the schoolmates' parents by one standard deviation determines an increase of 1.8% in lifetime earnings, with this effect being 3.6% for children from low socio-economic backgrounds. Moreover, the result highlight the importance of improved access to high-paying occupations due to network effects as a potential mechanism, with one student out of 5 being employed at the same plant where one of his schoolmates' parents was previously employed and exposure to schoolmates from higher earnings familes resulting in increased occupational status. Consistently with this fact, upon decomposing the mechanisms driving the exposure effect, I find that positive spillovers in human capital formation are offset by increased competition, but access to better jobs because of improved connections makes the effect positive. The evidence presented highlights the importance of considering the process of social mobility driven both by parental investments within the family and peer effects emergins from interactions among peers from different families, as members of schools and labor market networks.

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Data Source

Danish Administrative Data: DST https://www.dst.dk/da/

Tables

Table 1: Summary Statistics

		mean	sd	count
Individual Characteristics	Female	0.50	0.50	321,511
	Year of birth	1,983.33	2.08	321,511
	Age at enrollment	17.81	0.74	321,511
	School-Cohort size	290.21	227.18	321,511
	Earnings 28-32 (USD)	41,619.76	24,967.46	321,511
	Years of education	13.76	2.29	321,511
	College Degree	0.41	0.49	321,511
	Mother's earnings when kid 0-18	31,055.97	17,933.33	321,052
	Father's earnings when kid 0-18	54,922.99	38,539.05	316,781
	Age of mother (at birth)	27.54	4.79	321,128
	Age of father (at birth)	30.48	5.58	317,029
	College Degree, mother	0.24	0.43	321,511
	College Degree, father	0.21	0.40	321,511
Group Characteristics	Cohort	2,001.01	1.98	2,127
	Cohort size	151.16	145.01	2,127
	Avg. parental earnings (USD)	42,442.93	9,097.62	2,127
	Avg. parental earnings (percentiles)	50.44	11.55	2,127
	Students from $\geq$ 2 mun.	1.00	0.00	2,127
	Students from $\geq$ 2 par.	1.00	0.04	2,127

Table 2: Residual Variation in Schoolmates' Parental Earnings

	mean	sd	count
Schoolmates' Parental Earnings	50.94	9.94	321,511
Schoolmates' Parental Earnings - residual (linear trend)	0.00	2.21	321,511
Schoolmates' Parental Earnings - residual (nonlinear trend, 2nd order)	0.00	1.94	321,511
Schoolmates' Parental Earnings - residual (linear trend, 3rd order)	0.00	1.58	321,511
Schoolmates' Parental Earnings - residual (moving avg.)	0.00	2.28	144,069

Table 3: Parental Earnings are Orthogonal to (residuals of) Peers' Parental Earnings

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Average peers' parental earnings	0.044**	0.020	0.009	0.017	0.031	0.005	0.005	0.012
	(0.022)	(0.021)	(0.026)	(0.026)	(0.020)	(0.020)	(0.025)	(0.024)
Observations	321511	321511	321511	321511	316646	316646	316646	316646
School and time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Individual and school controls	No	No	No	No	Yes	Yes	Yes	Yes
School time trend	None	Linear	Quadratic	Cubic	None	Linear	Quadratic	Cubic
P-value of parental background	0.048	0.346	0.743	0.498	0.127	0.798	0.858	0.623

Note: The dependent variable in all columns is the percentile of parental earnings. All columns include the leave-one-out average of parenalt earnings at the school level, to account for negative mechanical bias due to using leave-one-out measure of peer characteristcs, as suggested by Guryan, Kroft, and Notowidigdo (2009) correction method. Individual controls included in Columns (5)-(8) include gender, year of birth and mother and father age at birth. School controls included in Columns (5)-(8) include the number of students enrolled in each school-cohort. The p-value reported in the last line refers to the coefficient on peers' parental earnings. All variables are standardized. SE clustered at the school level are reported in parenthesis. * p < 0.1, ** p < 0.05, *** p < 0.01.

Table 4: Predetermined characteristcs are Orthogonal to (residuals of) Peers' Parental Earnings

	N of test resulting in non-zero correlation			N of tests
	P-value<.1	P-value<.05	P-value<.01	
No time trend	26	23	17	32
	(81.25%)	(71.875%)	(53.125%)	(100%)
Linear time trend	5	4	1	32
	(15.625%)	(12.5%)	(3.125%)	(100%)

Note: This Table shows aggregate results from separate OLS regressions reported in Table 5. All regressions include cohort fixed effects and school fixed effects. The first row of the table refers to regressions which do not include school-specific time trends. The second row of the table refers to regressions which include school-specific time trends. The table reports the number (and the share in parentes) of variables which report correlation with the leave one out average of peers' parental earning different from zero at 90%, 95% and 99% condifence level.

Table 5: Predetermined characteristcs are Orthogonal to (residuals of) Peers' Parental Earnings

	No time trend			Linear Time trend		
	b	se	p	b	se	p
age entry	-0.031	(0.010)	0.002	-0.019	(0.006)	0.004
age father at birth	0.011	(0.005)	0.029	0.001	(0.004)	0.895
age mother at birth	0.018	(0.006)	0.002	0.000	(0.004)	0.950
mother teenager	-0.011	(0.005)	0.035	-0.004	(0.005)	0.404
father teenager	-0.009	(0.005)	0.104	-0.002	(0.005)	0.636
age father missing	-0.015	(0.005)	0.004	-0.003	(0.004)	0.426
age mother missing	-0.015	(0.006)	0.008	-0.007	(0.005)	0.132
years of education father	0.027	(0.006)	0.000	0.001	(0.005)	0.853
years of education mother	0.035	(0.007)	0.000	0.000	(0.004)	0.954
university degree father	0.022	(0.007)	0.001	-0.002	(0.005)	0.648
university degree mother	0.022	(0.007)	0.001	-0.002	(0.005)	0.648
education father missing	-0.015	(0.005)	0.004	-0.003	(0.004)	0.479
education mother missing	-0.016	(0.006)	0.005	-0.007	(0.004)	0.109
plant size [1, 49] father	0.016	(0.005)	0.001	0.008	(0.004)	0.043
plant size [1, 49] mother	0.010	(0.005)	0.034	0.002	(0.004)	0.623
plant size [50, 200] father	0.001	(0.005)	0.854	-0.004	(0.004)	0.369
plant size [50, 200] mother	0.011	(0.005)	0.022	0.003	(0.004)	0.480
plant size 200+ mother	0.006	(0.005)	0.216	-0.004	(0.004)	0.350
plant size 200+ father	0.006	(0.004)	0.203	-0.004	(0.004)	0.314
plant size father	0.004	(0.005)	0.427	0.000	(0.005)	0.968
plant size mother	-0.01	(0.005)	0.057	-0.010	(0.005)	0.030
Top 10% wage within plant, father	0.014	(0.006)	0.022	-0.001	(0.005)	0.791
Top 10% wage within plant, mother	0.009	(0.005)	0.075	0.001	(0.004)	0.743
Plant info missing father	-0.021	(0.006)	0.000	-0.002	(0.005)	0.643
Plant info missing mother	-0.024	(0.006)	0.000	-0.001	(0.004)	0.803
Female	-0.018	(0.006)	0.049	-0.011	(0.005)	0.037
Father ID missing	-0.015	(0.009)	0.006	-0.004	(0.004)	0.390
Mother ID missing	-0.017	(0.005)	0.005	-0.008	(0.005)	0.081
N of siblings	-0.018	(0.006)	0.005	-0.007	(0.005)	0.166
Birth order	-0.01	(0.006)	0.057	-0.005	(0.005)	0.275
First Born	0.002	(0.005)	0.754	0.003	(0.004)	0.459
Parish rank (by avg income) at birth	0.035	(0.011)	0.001	0.004	(0.006)	0.578

Table 6: Residuals in School Composition are Uncorrelated Over Time

	N of test resulting in non-zero correlation			N of tests
	P-value<.01	P-value<.05	P-value<.1	
None	3	17	14	332
	(0.9%)	(5.1%)	(4.2%)	(100%)
Linear	3	10	15	332
	(0.9%)	(3%)	(4.5%)	(100%)
Quadratic	3	12	12	332
	(0.9%)	(3.6%)	(3.6%)	(100%)
Cubic	4	5	9	332
	(1.2%)	(1.5%)	(2.7%)	(100%)

Note: This Table shows aggregate results from separate school-specific time series regressions. All regressions test for the school specific AR(1) coefficient of the correlation over time in school composition. School composition is measured as the average parental earning of students enrolled in each school and cohort. The first row of the table refers to regressions which do not include school-specific time trends. The latter rows of the table refers to regressions which include school-specific time trends. The table reports the number (and the share in parentes) of variables which report correlation with the leave one out average of peers' parental earning different from zero at 99%, 95% and 90% condifence level.

Table 7: Main Results

	(1)	(2)	(3)	(4)
Parental earnings, kid 0-18	0.203***	0.190***	0.184***	0.184***
	(0.006)	(0.005)	(0.005)	(0.005)
Schoomates' parental earnings		0.134***	0.179***	0.119**
		(0.022)	(0.036)	(0.039)
Observations	316646	316646	316646	316646
Cohort FE	No	No	Yes	Yes
School FE	No	No	Yes	Yes
School t trend	No	No	No	Yes
R^2	0.09	0.10	0.12	0.12

Note: Estimates from separate OLS regressions. dependent variable is children earnings by the age of 28-32. All earnings are expressed in ranks with respect tot the cohort-specific national distribution. All speifications include controls for year of birth, mother age at birth, father age at birth, gender and cohort size. SEs in parentheses are clustered at the school-cohort level. * p<0.10, ** p<0.05, *** p<0.01

Table 8: Higher Order Time Trends and Alternative Specifications

	(1)	(2)	(3)	(4)	(5)
Rank of avg parental earnings, kid 0-18	0.184***	0.183***	0.183***	0.194***	0.181***
	(0.005)	(0.005)	(0.005)	(0.007)	(0.005)
Rank of schoomates' parental earnings	0.119**	0.073*	0.080*		0.132**
	(0.039)	(0.038)	(0.038)		(0.037)
M.A. of rank schoomates' parental earnings				0.083*	
				(0.029)	
Observations	316646	316646	316646	141921	316431
School FE	Yes	Yes	Yes	No	Yes
Cohort FE	Yes	Yes	Yes	No	Yes
School t trend (1st order)	Yes	Yes	Yes	No	Yes
School t trend (2nd order)	No	Yes	Yes	No	No
School t trend (3rd order)	No	No	Yes	No	No
School×Municipality	No	No	No	No	Yes
R^2	0.12	0.13	0.13	0.10	0.13

Note: Estimates from separate OLS regressions. dependent variable is children earnings by the age of 28-32. All earnings are expressed in ranks with respect tot the cohort-specific national distribution. All speifications include controls for year of birth, mother age at birth, father age at birth, gender and cohort size. Municipality is defined as the municipality of recidence in the year of enrollment. SEs in parentheses are clustered at the school-cohort level. * p<0.10, ** p<0.05, *** p<0.01

Table 9: Placebo test

	(1)	(2)	(3)	(4)	(5)
$\tilde{X}_{i,t-2}$	0.010				
	(0.036)				
$\tilde{X}_{i,t-1}$		0.036			
		(0.029)			
$\bar{X}_{-i}$			0.119**		
			(0.039)		
$\tilde{X}_{i,t+1}$				-0.027	
				(0.038)	
$\tilde{X}_{i,t+2}$					-0.011
					(0.028)
Observations	236193	279426	316646	267730	220595

Note: Estimates from separate OLS regressions. dependent variable is children earnings by the age of 28-32. All earnings are expressed in ranks with respect tot the cohort-specific national distribution. All speifications include controls for year of birth, mother age at birth, father age at birth, gender and cohort size. $\tilde{X}_{i,t+c}$ is the average of parental earnings among students who enrolled in the same school as i, c years after the actual cohort of i. SEs in parentheses are clustered at the school-cohort level. * p<0.10, ** p<0.05, *** p<0.01

Table 10: Descriptives: Human capital and Employment

	Q1	Q2	Q3	Q4	Total
College Degree	0.280	0.345	0.436	0.573	0.410
	(0.449)	(0.475)	(0.496)	(0.495)	(0.429)
Firm FE	-0.0253	-0.0039	0.016	0.044	0.010
	(0.153)	(0.146)	(0.147)	(0.154)	(0.152)
Managerial Occ.	0.356	0.434	0.529	0.659	0.496
	(0.479)	(0.496)	(0.499)	(0.474)	(0.500)

Table 11: Descriptives: Network

Joined a plant where:	Q1	Q2	Q3	Q4	Total
Parent previously employed	0.031	0.057	0.064	0.083	0.059
	(0.173)	(0.232)	(0.276)	(0.276)	(0.236)
Schoolmate employed	0.163	0.180	0.179	0.172	0.174
	(0.370)	(0.384)	(0.383)	(0.378)	(0.379)
Schoolmate (high-SES) employed	0.046	0.054	0.062	0.081	0.061
	(0.210)	(0.226)	(0.242)	(0.273)	(0.240)
Schoolmate's parent previously employed	0.276	0.305	0.294	0.272	0.287
	(0.447)	(0.460)	(0.456)	(0.445)	(0.452)
Schoolmate's (high-SES) parent previously employed	0.079	0.093	0.110	0.137	0.105
	(0.270)	(0.290)	(0.311)	(0.344)	(0.306)

Table 12: GMM Parameters

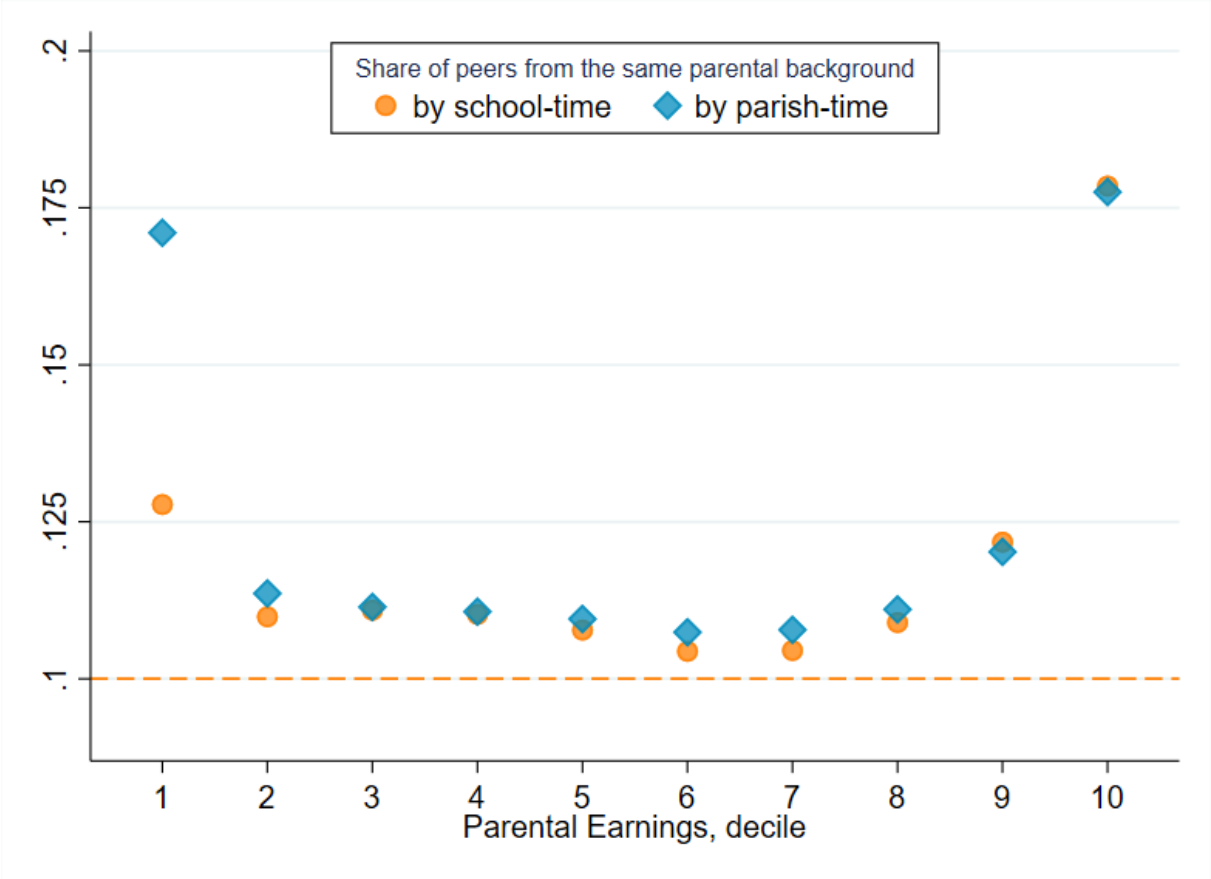
δ	0.011	(0.000)
$\frac{\beta\delta+\gamma}{1-\beta}$	0.012	(0.002)
$\frac{\delta+\gamma}{1-\beta}$	0.025	(0.000)
π	9.39	(0.585)
σ	-6.556	(3.079)
β'	0.104	(0.051)
ρ	0.537	(0.011)
λ	-0.013	(0.013)
ϕ	0.223	(0.155)
μ_1	0.448	(0.006)
ω	0.238	(0.035)

Table 13: Model Parameters

Spillovers in school	$\frac{\beta\delta+\gamma}{1-\beta}\pi$	0.112	(0.02)
Spillovers after school	$\frac{\sigma}{1-\beta'}\frac{\delta+\gamma}{1-\beta}$	-0.185	(0.086)
Network	$\frac{\delta+\gamma}{1-\beta}\pi\frac{\beta'}{1-\beta'}$	0.028	(0.015)
Parental Network	$\mu_1\left(\frac{\beta'}{1-\beta'}\lambda+\frac{\phi}{1-\beta'}\right)$	0.111	(0.078)
Total	ψ_2	0.073	(0.030)

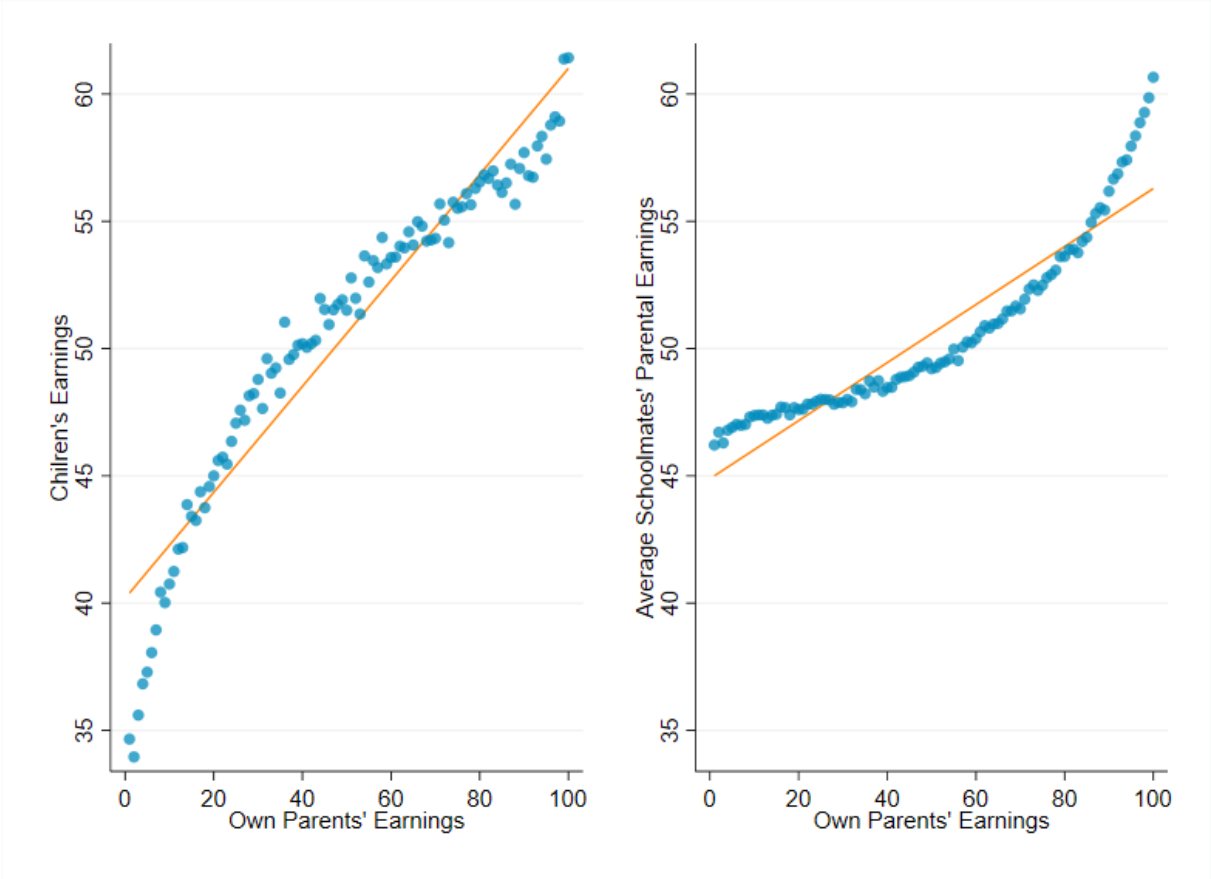
Figures

Figure 1: Sorting across Schools and Neighborhoods



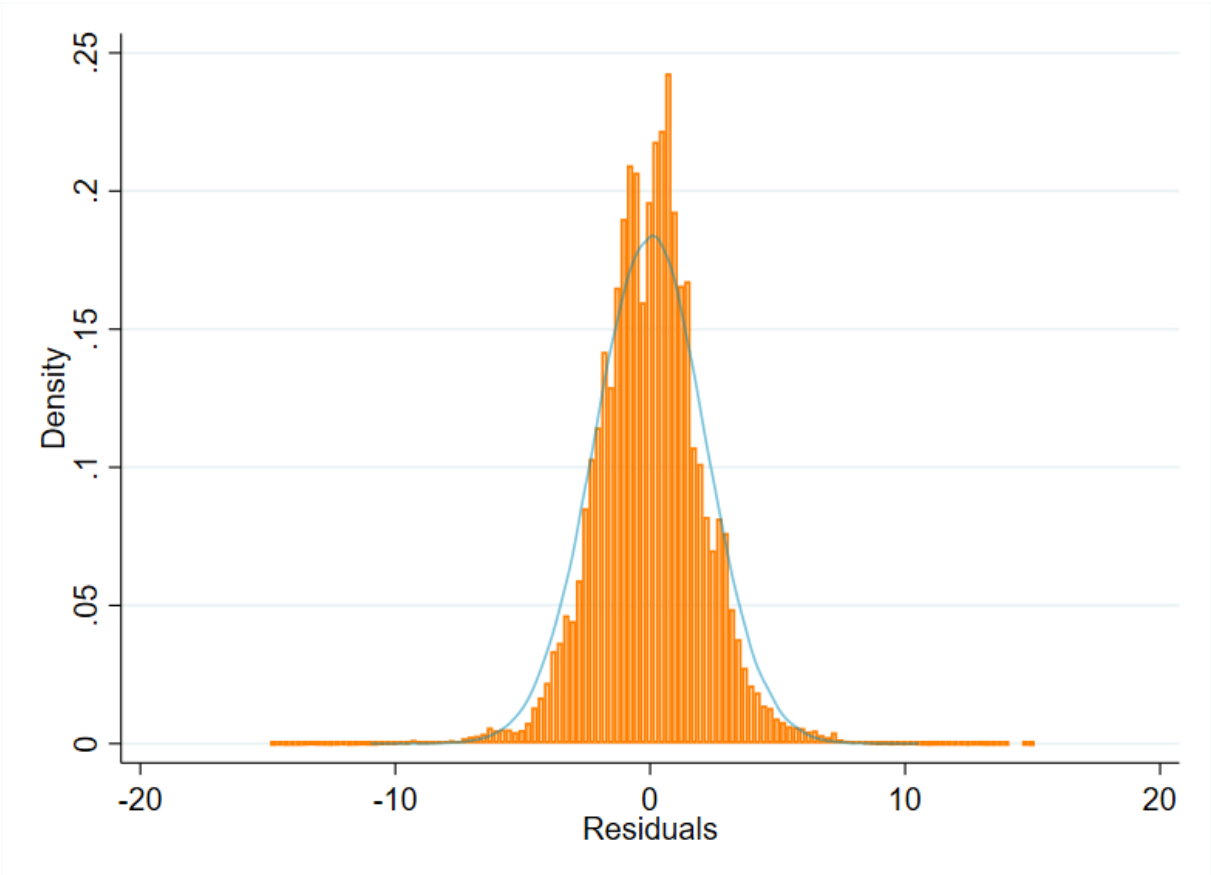
The graph plots the share of peers from the same parental background, conditional on the decile of the parental earnings distribution. Orange dots represent the probability a given schoolmate's parents are from the same decile of the parental earnings distribution for each decile of the parental earnings distribution. Blue diamonds represent the probability a given neighbor's parents are from the same decile of the parental earnings distribution for each decile of the parental earnings distribution. Neighbors are defined as individuals born in the same calendar year and registered as living in the same parish for that year.

Figure 2: Intergenerational Mobility and Peer Exposure



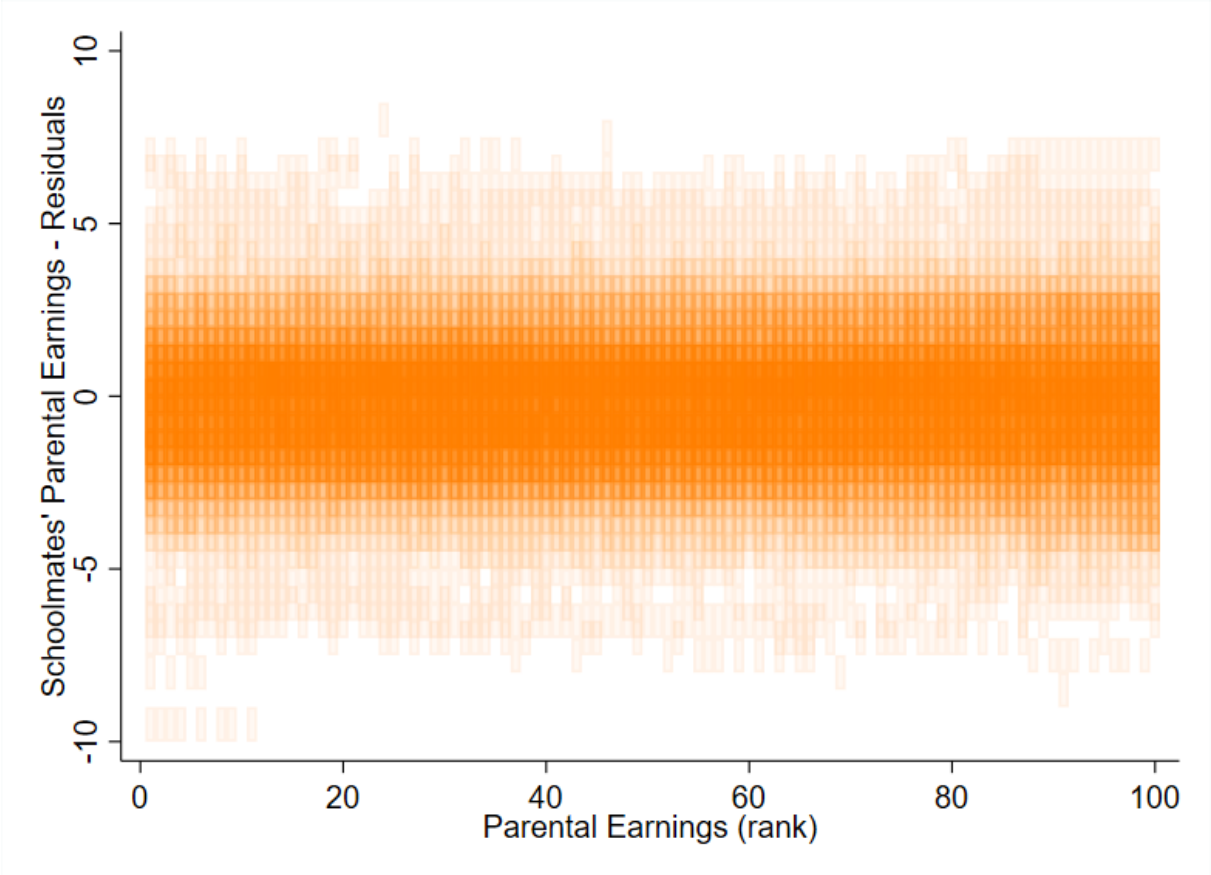
The left graph plots the average earnings by the age 28-32 conditional on own parental earnings. Both earnings and parental earnings are measured in percentiles of the national earnings distribution. The right graph plots the average peers' parental earnings conditional on own parental earnings. Peers' parental earnings are computed as the leave one out average of parental earnings among schoolmates.

Figure 3: Residuals are Normally Distributed



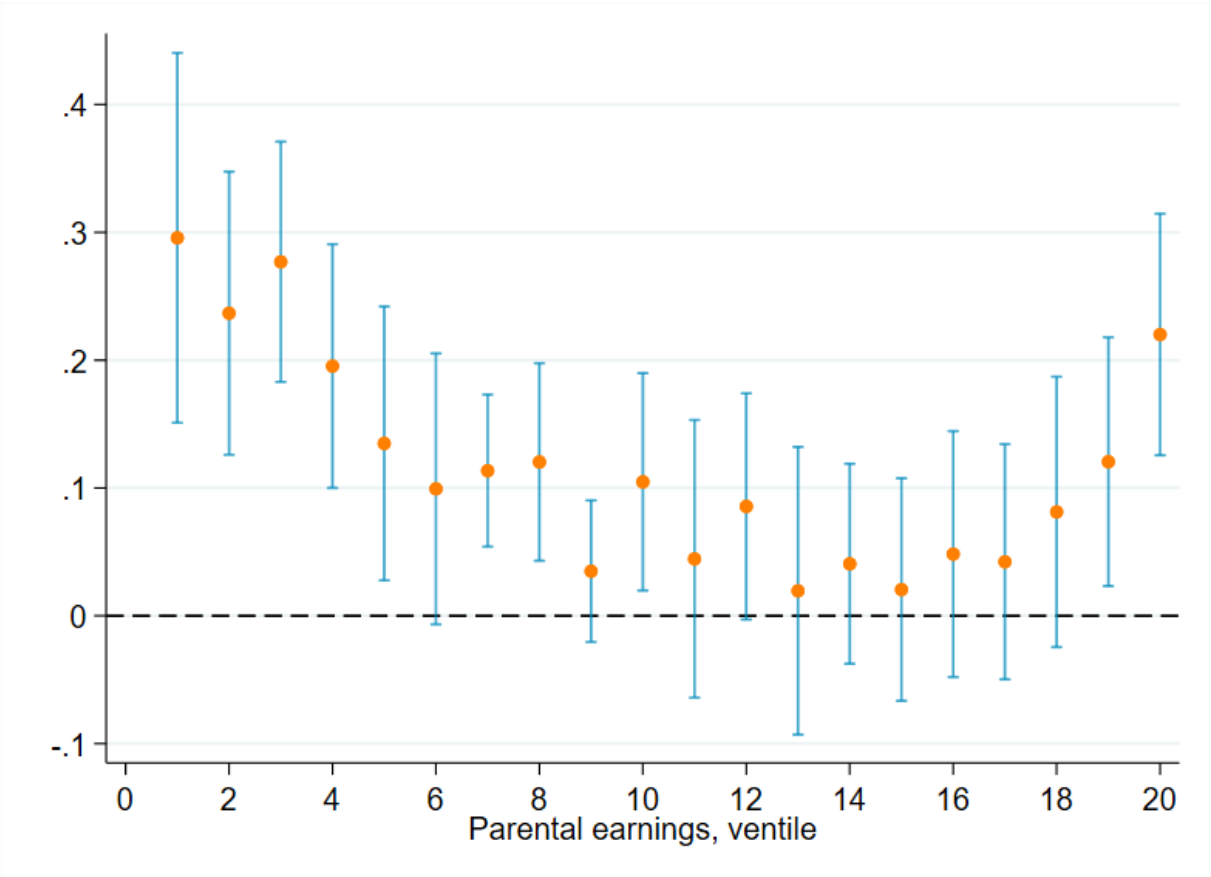
The graph plots the empirical PDF of the residual of average schoolmates' parental background from a regression on school fixed effects and school specific linear time trends. Average schoolmates' parental background is defines as the leave-one-out average of schoolmates parental earnings in percentiles of the national distribution. The PDF of a normal distribution with the same mean and standard deviation as the resuduals is represented by the continous blue line.

Figure 4: Parental Earnings are Orthogonal to (residuals of) Peers' Parental Earnings



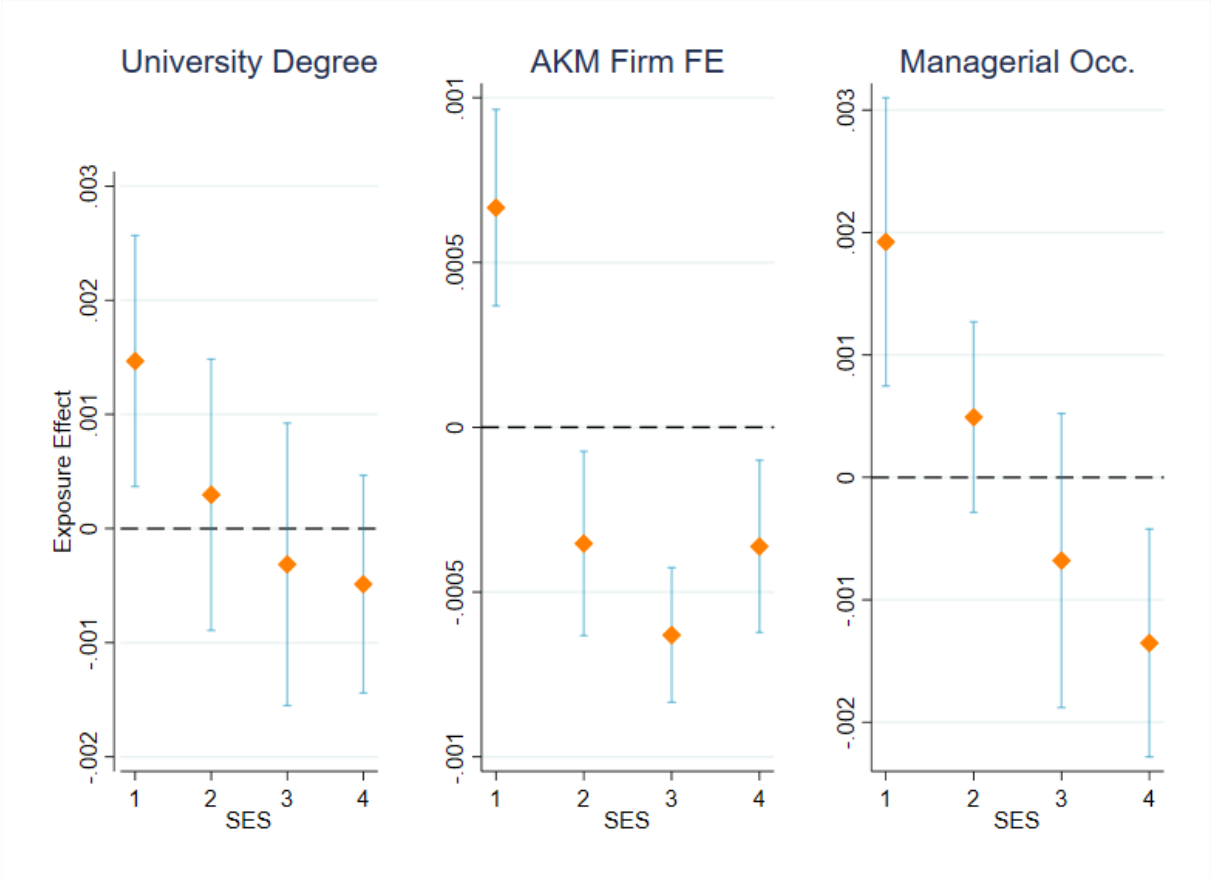
The graph plots the empirical bivariate distribution of the residual of average schoolmates' parental background from a regression on school fixed effects (vertical axis) and school specific linear time trends and own parental background (horizontal axis). Average schoolmates' parental background is defines as the leave-one-out average of schoolmates parental earnings in percentiles of the national distribution.

Figure 5: Heterogeneity by SES



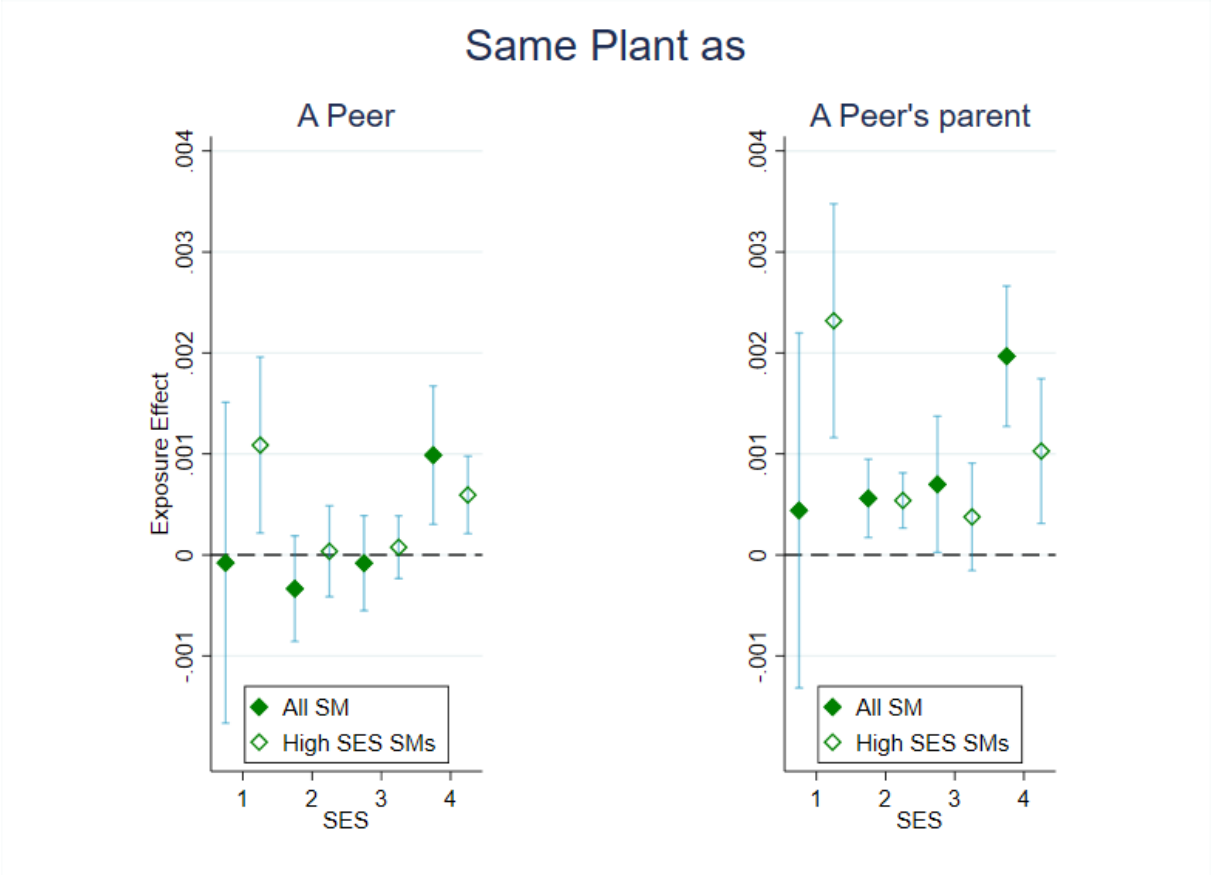
The graph plots the marginal effect (along with 95% confidence intervals) of the coefficient measuring the impact of peers' parental earnings on own earnings from a version of eq. 1 which includes a full set of interactions between the variable of interest and a set of dummies for each ventile of parental earnings.

Figure 6: Exposure Effects: Human Capital and Labor Market Outcomes



The graph plots the marginal effect (along with 95% confidence intervals) of the coefficient measuring the impact of peers' parental earnings on the dependent vairable from a version of eq. 1 which includes a full set of interactions between the variable of interest and a set of dummies for each quartile of parental earnings. In the left panel the depent variable is the probability of obtaining a college degree, in the middle panel is the AKM firm fixed-effect at the firm of employement at the age of 30 and in the right panel is the probability ob being employed in an occupation with managerial tasks.

Figure 7: Exposure Effects: Network



The graph plots the marginal effect (along with 95% confidence intervals) of the coefficient measuring the impact of peers' parental earnings on the dependent vairable from a version of eq. 1 which includes a full set of interactions between the variable of interest and a set of dummies for each quartile of parental earnings. In the left panel the depent variable is the probability of joining the plant where a schoolmate or a high SES schoolmate is employed. In the right panel the depent variable is the probability of joining the plant where a schoolmate's parent or a high SES schoolmate's parent was previously employed.

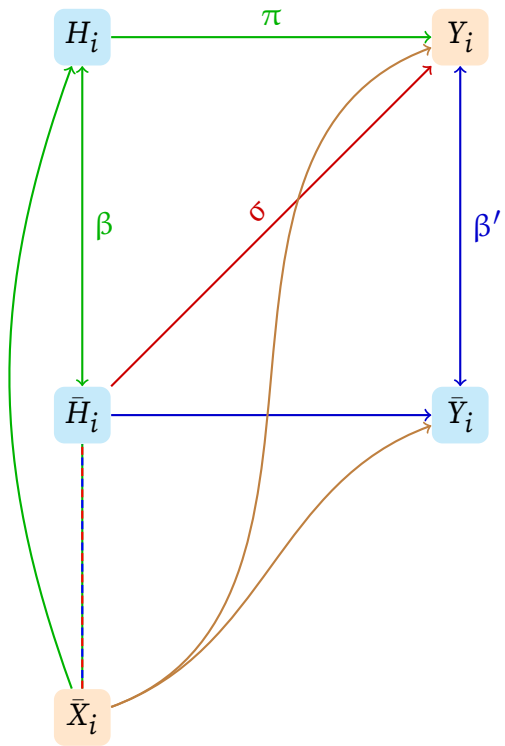


Figure 8: A Graphical Representation of the Model

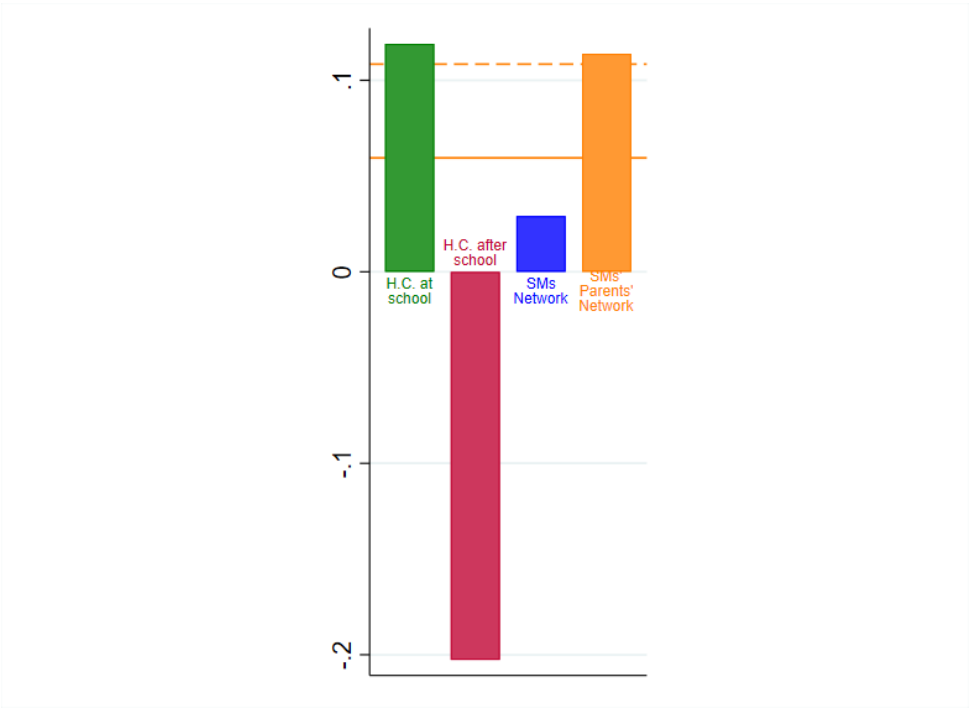


Figure 9

Extra

Table 14

	(1)
quartile=1 × Rank of avg parental earnings, kid 0-18	0.427***
	(0.019)
quartile=2 × Rank of avg parental earnings, kid 0-18	0.169***
	(0.016)
quartile=3 × Rank of avg parental earnings, kid 0-18	0.119***
	(0.018)
quartile=4 × Rank of avg parental earnings, kid 0-18	0.136***
	(0.020)
Rank of schoomates' parental earnings	0.227***
	(0.052)
quartile=2	12.819***
	(1.296)
quartile=3	18.307***
	(1.354)
quartile=4	13.411***
	(2.421)
quartile=2 × Rank of schoomates' parental earnings	-0.131***
	(0.031)
quartile=3 × Rank of schoomates' parental earnings	-0.185***
	(0.027)
quartile=4 × Rank of schoomates' parental earnings	-0.123***
	(0.031)
female	-13.144***
	(0.437)
father year of birth	0.035
	(0.020)
mother year of birth	0.038
	(0.023)
kid year of birth	5.004***
	(0.286)
group_size	-0.004**
	(0.001)
Constant	-10032.833***
	(564.755)
Observations	316646
R^2	0.13

SEs in parentheses are clustered at the school-cohort level.

Cohort, municipality, gender, childrens' and parents' yr birth FEs are included.

School FEs and school-specific time trends are also included.

* p<0.10, ** p<0.05, *** p<0.01

Table 15

	College Degree		Firm FE		Managerial Position	
	(1)	(2)	(3)	(4)	(5)	(6)
barX	0.0013**	0.0013**	0.0004**	0.0007***	0.0015**	0.0019***
	(0.0005)	(0.0004)	(0.0001)	(0.0001)	(0.0005)	(0.0005)
barX x q2		0.0003		-0.0003**		0.0005
		(0.0005)		(0.0001)		(0.0003)
barX x q3		-0.0003		-0.0006***		-0.0007
		(0.0005)		(0.0001)		(0.0005)
barX x q4		-0.0005		-0.0004**		-0.0014**
		(0.0004)		(0.0001)		(0.0004)
X	0.0018***	0.0012***	0.0007***	0.0010***	0.0022***	0.0024***
	(0.0000)	(0.0002)	(0.0000)	(0.0001)	(0.0001)	(0.0002)
group_size	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
X x q2		0.0005*		-0.0005***		-0.0002
		(0.0002)		(0.0001)		(0.0003)
X x q3		0.0007		-0.0004**		-0.0001
		(0.0004)		(0.0001)		(0.0004)
X x q4		0.0010**		0.0000		-0.0008
		(0.0004)		(0.0001)		(0.0004)
q2		-0.0236		0.0299***		-0.0236
		(0.0261)		(0.0072)		(0.0126)
q3		-0.0028		0.0363***		0.0238
		(0.0424)		(0.0070)		(0.0268)
q4		-0.0119		-0.0091		0.1222***
		(0.0301)		(0.0080)		(0.0322)
N	316646	316646	266169	266169	316646	316646
Sample Mean	.41		.009		.496	
S. Mean 1q		.28		-.025		.356
R^2	0.27	0.27	0.08	0.08	0.20	0.20

time, municipality, gender, childrens' and parents' yr birth FEs are included.

SEs in parentheses are clustered at the school-time level.

* p<0.10, ** p<0.05, *** p<0.01

Table 16

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Join SM		Join high SES SM		Join SM par		Join high SES SM par	
barX	0.0001	-0.0001	0.0013***	0.0011**	0.0013	0.0004	0.0028***	0.0023***
	(0.0006)	(0.0006)	(0.0003)	(0.0004)	(0.0007)	(0.0007)	(0.0004)	(0.0005)
barX x q2		-0.0003		0.0000		0.0006**		0.0005***
		(0.0002)		(0.0002)		(0.0002)		(0.0001)
barX x q3		-0.0001		0.0001		0.0007**		0.0004
		(0.0002)		(0.0001)		(0.0003)		(0.0002)
barX x q4		0.0010**		0.0006***		0.0020***		0.0010**
		(0.0003)		(0.0002)		(0.0003)		(0.0003)
X	0.0002***	0.0008***	0.0002***	0.0003**	0.0000	0.0015***	0.0002***	0.0006***
	(0.0000)	(0.0002)	(0.0000)	(0.0001)	(0.0000)	(0.0002)	(0.0000)	(0.0001)
group_size	0.0002***	0.0002***	0.0001***	0.0001***	0.0003***	0.0003***	0.0001***	0.0001***
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
X x q2		-0.0005*		-0.0002		-0.0013**		-0.0005**
		(0.0002)		(0.0001)		(0.0004)		(0.0002)
X x q3		-0.0008**		-0.0002		-0.0014***		-0.0002
		(0.0003)		(0.0002)		(0.0003)		(0.0001)
X x q4		-0.0012***		-0.0002		-0.0027***		-0.0008***
		(0.0002)		(0.0001)		(0.0003)		(0.0002)
q2		0.0302**		0.0057		0.0110		-0.0101
		(0.0118)		(0.0090)		(0.0128)		(0.0074)
q3		0.0298		-0.0003		-0.0007		-0.0202
		(0.0163)		(0.0086)		(0.0156)		(0.0129)
q4		0.0060		-0.0200*		0.0291		-0.0134
		(0.0197)		(0.0094)		(0.0245)		(0.0211)
N	316630	316630	316630	316630	316630	316630	316630	316630
Sample Mean	.172		.061		.285		.104	
S. Mean 1q		.163		.046		.276		.079
R^2	0.03	0.03	0.02	0.02	0.04	0.04	0.05	0.05

time, municipality, gender, childrens' and parents' yr birth FEs are included.

SEs in parentheses are clustered at the school-time level.

* p<0.10, ** p<0.05, *** p<0.01